Midterm Exam for “The Mathematical Foundations of Political Methodology”

You have our class time for this exam. You may use a calculator and refer to your Pemberton & Rau textbook. Please do not use the internet during the exam. Show your work and be tidy!

1. Given the following hold, what (if anything) do we know about the relationship between sets A and C?

   (a) $A \cap B = A$ and $B \cap C \neq \emptyset$

   (b) $A \cup B = A$ and $B \cap C \neq \emptyset$

   (c) $C \cup D = B$ and $B \cap C = A$

2. Find the limit of each of the following sequences as $n \to \infty$, if the limit exists.

   (a)
   $$\frac{4n^3 - 1}{(n+1)^3}$$

   (b)
   $$\frac{(-1)^n n^{\frac{5}{8}}}{7 + n^{\frac{7}{8}}}$$

   (c)
   $$\frac{(-1)^n n^{\frac{1}{2}}}{n^{\frac{1}{2}} + n^{\frac{2}{5}}}$$

3. Use the (δ, ϵ) definition of a limit to prove that the function
   $$f(x) = 1/2x - 10$$
   is everywhere continuous.

4. Let
   $$f(x) = x^2/(x+4)$$

   (a) Find the critical points of f and characterize them as maxima, minima, or inflection points.

   (b) Find the regions on which f is increasing and decreasing.

   (c) Find any asymptotes that f may have.

   (d) Use your answers above to plot out f.

5. Perform the following matrix operations:

   (a) Find the inverse of

$$\begin{bmatrix} 1 & 1 & 4 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

   (b) Find the rank of

$$\begin{bmatrix} 1 & 1 & 5 \\ 0 & 2 & 4 \\ -2 & 0 & -6 \\ 1 & 9 & 21 \end{bmatrix}$$

   (c) Check whether the following matrix is positive definite, positive semidefinite, negative definite, negative semidefinite, or none of the above.

$$\begin{bmatrix} 1 & 1 & 2 \\ 1 & 0 & 1 \\ 2 & 1 & 0 \end{bmatrix}$$

6. Find and classify all of the critical points of the following functions:

   (a)

$$f(x, y) = -10x^2 + 4xy - \frac{2y^2}{5}$$

   (b)

$$f(x, y, z) = -x^2 + \log(x) - \frac{1}{y} - y - z^2 + z \quad \text{for } x, y, z > 0$$

   (c)

$$f(x, y) = x^2 + y^2 - xy \text{ subject to the constraint that } 2x - y = 3$$